

TrES-3b

R-1995

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130517 · created 2026-08-07T15:32:06+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456429.9379 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.015
Transit depth (Rp/Rs)^2	3.14 +/- 0.53 [%]
Orbital Inclination (inc)	82.76 +/- 0.63
Airmass coefficient 1 (a1)	0.954 +/- 0.019
Airmass coefficient 2 (a2)	0.04 +/- 0.1
Scatter in the residuals of the lightcurve fit is	1.94 %
Best Comparison Star	#2 - [517, 163]
Optimal Aperture	2.72
Optimal Annulus	11.65
Transit Duration (day)	0.0636 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.015 vs 0.16309 (deviation 0.76σ)
Duration fit vs published	0.0636 d vs 0.0567 d (deviation 1.23σ)
Mid-time O–C	-2.5 min
Depth SNR	9.6
Residual scatter	1.94 %

Light curve

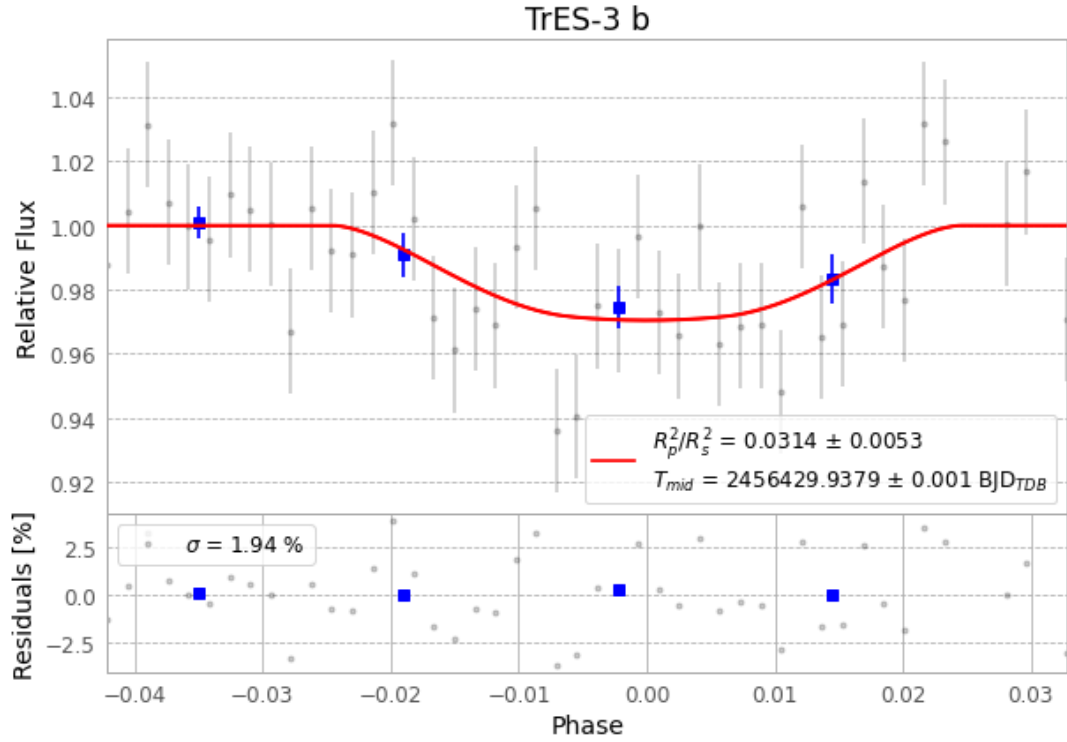


Figure 1 — FinalLightCurve_TrES-3 b_2013-05-17.png (SHA-256 06f0405468b81c7e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [312, 272], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ff84fa855c462fd2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

48 FITS frames (32 MB), TRES-3130517090612.FITS → TRES-3130517112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130517.log (8cb587d88d1e...), FinalLightCurve_TrES-3 b_2013-05-17.png (06f0405468b8...), FinalLightCurve_TrES-3 b_2013-05-17.csv (a5c16407fc7e...)

Compute resources

Wall time 323.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 15:32Z.